

DEPT:	CSE, IT and Allied Departments			
SEMESTER:	5th			
SUBJECT CODE:	PCCCS503			
SUBJECT NAME:	Operating Systems			
Module No.	Text Book Name, Author, Publisher	Section No. (if any)	Chapter No	Exercise Nos.
Module 1	Operating System Principles, Silberschatz, Galvin and Gagne, Wiley India Edition		1, 2,3,5	Chapter1: 1.1,1.2,1.3,1.4,1.5,1.6,1.7,1.8,1.9,1.10,1.11,1.12 Chapter 2: 2.1,2.2,2.3,2.4,2.5,2.6 Chapter 3: 3.1,3.2,3.3,3.4 Chapter 5: 5.1,5.2,5.3,5.4,5.5,5.6,5.7,5.8
Module 2	Operating System Principles, Silberschatz, Galvin and Gagne, Wiley India Edition		6,7	Chapter 6: 6.1,6.2,6.3,6.4 Chapter 7: 7.5,7.6,7.7,7.8
Module 2	Operating Systems, William Stallings, Pearson Education		5,6	Chapter 5: 5.4,5.5,5.6,5.7,5.8,5.9,5.10,5.11,5.12,5.13 Chapter 6: 6.1,6.2,6.3,6.4,6.5,6.6,6.7
Module 3	Operating Systems, William Stallings,		7,8	Chapter 7: Review Question: 7.1,7.2,7.3,7.4,7.5,7.6,7.7,7.8,7.9 Problem:

	Pearson Education			7.1,7.2,7.3,7.4,7.5,7.6,7.7,7.8,7.9,7.10 ,7.11,7.12 Chapter 8: 8.5,8.6,8.7
Module 4	Operating Systems, William Stallings, Pearson Education		12	Chapter 12: 12.1,12.2,12.3,12.4,12.5,12.6,12.7,12. 8,12.9,12.10,12.11
1. Link for Study Material: https://drive.google.com/file/d/1sbcOy8Q779CIReZhqZldMaMolLYvAYr7/view?usp=sharing				

Subject	Operating System
Module	I
Chapters	1,2,3,5
Author	Galvin

Exercises

- 1.1 In a multiprogramming and time-sharing environment, several users share the system simultaneously. This situation can result in various security problems.
 - a. What are two such problems?
 - b. Can we ensure the same degree of security in a time-shared machine as in a dedicated machine? Explain your answer.
- 1.2 The issue of resource utilization shows up in different forms in different types of operating systems. List what resources must be managed carefully in the following settings:
 - a. Mainframe or minicomputer systems
 - b. Workstations connected to servers
 - c. Handheld computers
- 1.3 Which of the functionalities listed below need to be supported by the operating system for the following two settings: (a) handheld devices and (b) real-time systems.
 - a. Batch programming
 - b. Virtual memory
 - c. Time sharing
- 1.4 Describe the differences between symmetric and asymmetric multiprocessing. What are three advantages and one disadvantage of multiprocessor systems?
- 1.5 Distinguish between the client-server and peer-to-peer models of distributed systems.
- 1.6 Consider a computing cluster consisting of two nodes running a database. Describe two ways in which the cluster software can manage access to the data on the disk. Discuss the benefits and disadvantages of each.
- 1.7 What is the purpose of interrupts? What are the differences between a trap and an interrupt? Can traps be generated intentionally by a user program? If so, for what purpose?
- 1.8 Direct memory access is used for high-speed I/O devices in order to avoid increasing the CPU's execution load.
 - a. How does the CPU interface with the device to coordinate the transfer?
 - b. How does the CPU know when the memory operations are complete?
 - c. The CPU is allowed to execute other programs while the DMA controller is transferring data. Does this process interfere with

the execution of the user programs? If so, describe what forms of interference are caused.

- 1.9 Give two reasons why caches are useful. What problems do they solve? What problems do they cause? If a cache can be made as large as the device for which it is caching (for instance, a cache as large as a disk), why not make it that large and eliminate the device?
- 1.10 Discuss, with examples, how the problem of maintaining coherence of cached data manifests itself in the following processing environments:
- a. Single-processor systems
 - b. Multiprocessor systems
 - c. Distributed systems
- 1.11 What network configuration would best suit the following environments?
- a. A dormitory floor
 - b. A university campus
 - c. A state
 - d. A nation
- 1.12 Define the essential properties of the following types of operating systems:
- a. Batch
 - b. Interactive
 - c. Time sharing
 - d. Real time
 - e. Network
 - f. Parallel
 - g. Distributed
 - h. Clustered
 - i. Handheld

Exercises

- 2.1** The services and functions provided by an operating system can be divided into two main categories. Briefly describe the two categories and discuss how they differ.
- 2.2** List five services provided by an operating system that are designed to make it more convenient for users to use the computer system. In what cases it would be impossible for user-level programs to provide these services? Explain.

importance of obtaining such a statistical profile.

- 2.4 What are the five major activities of an operating system with regard to file management?
- 2.5 What is the purpose of the command interpreter? Why is it usually separate from the kernel? Would it be possible for the user to develop a new command interpreter using the system-call interface provided by the operating system?
- 2.6 What are the two models of interprocess communication? What are the strengths and weaknesses of the two approaches?

Exercises

- 3.1 Describe the differences among short-term, medium-term, and long-term scheduling.
- 3.2 Describe the actions taken by a kernel to context-switch between processes.
- 3.3 Using the program shown in Figure 3.23, explain what will be output at Line A.
- 3.4 What are the benefits and the disadvantages of each of the following? Consider both the system level and the programmer level.
 - a. Synchronous and asynchronous communication
 - b. Automatic and explicit buffering
 - c. Send by copy and send by reference
 - d. Fixed-sized and variable-sized messages

ercises

5.1 Discuss how the following pairs of scheduling criteria conflict in certain settings.

- a. CPU utilization and response time
- b. Average turnaround time and maximum waiting time
- c. I/O device utilization and CPU utilization

5.2 Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process	Burst Time	Priority
P_1	10	3
P_2	1	1
P_3	2	3
P_4	1	4
P_5	5	2

The processes are assumed to have arrived in the order P_1, P_2, P_3, P_4, P_5 , all at time 0.

- a. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, nonpreemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1).
 - b. What is the turnaround time of each process for each of the scheduling algorithms in part a?
 - c. What is the waiting time of each process for each of the scheduling algorithms in part a?
 - d. Which of the algorithms in part a results in the minimum average waiting time (over all processes)?
- 5.3 Why is it important for the scheduler to distinguish I/O-bound programs from CPU-bound programs?
- 5.4 Which of the following scheduling algorithms could result in starvation?
- a. First-come, first-served
 - b. Shortest job first
 - c. Round robin
 - d. Priority
- 5.5 Consider a system running ten I/O-bound tasks and one CPU-bound task. Assume that the I/O-bound tasks issue an I/O operation once for every millisecond of CPU computing and that each I/O operation takes 10 milliseconds to complete. Also assume that the context-switching overhead is 0.1 millisecond and that all processes are long-running tasks. What is the CPU utilization for a round-robin scheduler when:
- a. The time quantum is 1 millisecond
 - b. The time quantum is 10 milliseconds
- 5.6 Consider a system implementing multilevel queue scheduling. What strategy can a computer user employ to maximize the amount of CPU time allocated to the user's process?
- 5.7 Explain the differences in the degree to which the following scheduling algorithms discriminate in favor of short processes:
- a. FCFS
 - b. RR
 - c. Multilevel feedback queues
- 5.8 Using the Windows XP scheduling algorithm, what is the numeric priority of a thread for the following scenarios?
- a. A thread in the `REALTIME_PRIORITY_CLASS` with a relative priority of `HIGHEST`

Study Material Questions for Module I

Q.No.	Question	Blooms Level	CO	Marks						
1	There are three jobs running in a multi-programming environment with the following requirements: Job1: Requires disk after every 2 min (device service time including wait and access= 2 min). Total processing time= 6 min. Job2: Requires printer after every 5 min (device service time including wait and access= 2 min). Total processing time= 7 min. Job3: Requires disk after every 3 min (device service time including wait and access= 2min). Total processing time= 5 min. Prepare a timing chart showing the CPU and I/O activities of the jobs. Compute the total time for execution using mono-programming and multi-programming and then compare the results.	4	1	10						
2	What event handler would be executed in the following cases: a. The running process has finished its execution before completion of its time slice. b. The running process tries to access memory location that is not allowed to access. c. If there is failure in reading or writing an I/O device. d. A process is ready to execute but there is no space in the main memory. e. A periodic process is idle waiting for its next time slot to be executed.	4	1	10						
3	Processes go through the following states in their lifetime. Process Life Cycle <pre>graph TD new((new)) --> admitted((admitted)) admitted --> ready((ready)) ready -- "scheduler dispatch" --> running((running)) running -- "I/O or event wait" --> waiting((waiting)) waiting -- "I/O or event completion" --> ready ready -- "I/O or event completion" --> ready running -- "exit" --> terminated((terminated))</pre> Schedulers: manage queues Consider the following events and answer the questions that follow. Assume there are 5 processes, all either in the read or running states initially. Assume the processes are using a single processor. • At time 5: P1 executes a command to read from disk 3. • At time 15: P3's time slice ends. • At time 18: P4 executes a command to write to disk 3. • At time 20: P2 executes a command to read from disk 2. • At time 24: P3 executes a command to join with P5. • At time 33: An interrupt occurs indicating that P2's read is complete. • At time 36: An interrupt occurs indicating that P1's read is complete. • At time 38: P5 terminates. • At time 48: An interrupt occurs indicating that P4's write is complete. For time 22, 37 and 47, identify which state each process is in. If it is waiting, indicate what it is waiting for.	6	1	10						
4	Consider the 3 processes, P1, P2 and P3 shown in the table. <table><tr><th>Process</th><th>Arrival time</th><th>Time Units Required</th></tr><tr><td>P1</td><td>0</td><td>5</td></tr></table>	Process	Arrival time	Time Units Required	P1	0	5	4	1	10
Process	Arrival time	Time Units Required								
P1	0	5								

	P217 P334 What will be the completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units)?															
5	Three processes A, B and C each execute a loop of 100 iterations. In each iteration of the loop, a process performs a single computation that requires t_c CPU milliseconds and then initiates a single I/O operation that lasts for t_{io} milliseconds. It is assumed that the computer where the processes execute has sufficient number of I/O devices and the OS of the computer assigns different I/O devices to each process. Also, the scheduling overhead of the OS is negligible. The processes have the following characteristics: <table><tr><td>Process id</td><td>t_c</td><td>t_{io}</td></tr><tr><td>A</td><td>100ms</td><td>500 ms</td></tr><tr><td>B</td><td>350 ms</td><td>500 ms</td></tr><tr><td>C</td><td>200 ms</td><td>500 ms</td></tr></table> The processes A, B, and C are started at times 0, 5 and 10 milliseconds respectively, in a pure time sharing system (round robin scheduling) that uses a time slice of 50 milliseconds. What is the time in milliseconds at which process C would complete its first I/O operation?	Process id	t_c	t_{io}	A	100ms	500 ms	B	350 ms	500 ms	C	200 ms	500 ms	6	1	10
Process id	t_c	t_{io}														
A	100ms	500 ms														
B	350 ms	500 ms														
C	200 ms	500 ms														
6	Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.	5	1	10												
7	Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?	6	1	10												
8	Consider the following table of arrival time and burst time for three processes P0, P1 and P2. <table><tr><td>Process</td><td>Arrival time</td><td>Burst Time</td></tr><tr><td>P0</td><td>0 ms</td><td>9 ms</td></tr><tr><td>P1</td><td>1 ms</td><td>4 ms</td></tr><tr><td>P2</td><td>2 ms</td><td>9 ms</td></tr></table> The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?	Process	Arrival time	Burst Time	P0	0 ms	9 ms	P1	1 ms	4 ms	P2	2 ms	9 ms	4	1	10
Process	Arrival time	Burst Time														
P0	0 ms	9 ms														
P1	1 ms	4 ms														
P2	2 ms	9 ms														
9	Assume every process requires 3 seconds of service time in a system with single processor. If new processes are arriving at the rate of 10 processes per minute, then estimate the fraction of time CPU is busy in system?	4	1	10												
10	Under what circumstances does a multithreaded solution using multiple kernel threads provide better performance than a single- threaded solution on a single-processor system? Describe the actions taken by a thread library to context switch between user- level threads.	5	1	10												

Subject	Operating System
Module	II
Chapters	6,7
Author	Galvin

Exercises

- 6.1** The first known correct software solution to the critical-section problem for two processes was developed by Dekker. The two processes, P_0 and P_1 , share the following variables:

```
boolean flag[2]; /* initially false */  
int turn;
```

The structure of process P_i ($i == 0$ or 1) is shown in Figure 6.27; the other process is P_j ($j == 1$ or 0). Prove that the algorithm satisfies all three requirements for the critical-section problem.

- 6.2** Explain why spinlocks are not appropriate for single-processor systems yet are often used in multiprocessor systems.
- 6.3** Explain why implementing synchronization primitives by disabling interrupts is not appropriate in a single-processor system if the synchronization primitives are to be used in user-level programs.
- 6.4** Describe how the `Swap()` instruction can be used to provide mutual exclusion that satisfies the bounded-waiting requirement.

... decrease the number of processes.

- 7.5** Consider a system consisting of m resources of the same type being shared by n processes. Resources can be requested and released by processes only one at a time. Show that the system is deadlock free if the following two conditions hold:
- The maximum need of each process is between 1 and m resources.
 - The sum of all maximum needs is less than $m + n$.
- 7.6** Consider the dining-philosophers problem where the chopsticks are placed at the center of the table and any two of them could be used by a philosopher. Assume that requests for chopsticks are made one at a time. Describe a simple rule for determining whether a particular request could be satisfied without causing deadlock given the current allocation of chopsticks to philosophers.
- 7.7** We can obtain the banker's algorithm for a single resource type from the general banker's algorithm simply by reducing the dimensionality of the various arrays by 1. Show through an example that the multiple-resource-type banker's scheme cannot be implemented by individual application of the single-resource-type scheme to each resource type.
- 7.8** Consider the following snapshot of a system:

	<u>Allocation</u>	<u>Max</u>	<u>Available</u>
	<i>A B C D</i>	<i>A B C D</i>	<i>A B C D</i>
P_0	0 0 1 2	0 0 1 2	1 5 2 0
P_1	1 0 0 0	1 7 5 0	
P_2	1 3 5 4	2 3 5 6	
P_3	0 6 3 2	0 6 5 2	
P_4	0 0 1 4	0 6 5 6	

Answer the following questions using the banker's algorithm:

- What is the content of the matrix *Need*?
- Is the system in a safe state?
- If a request from process P_1 arrives for (0,4,2,0), can the request be granted immediately?

Module	Topic	Book
Module-2	Process Synchronization	Operating System- William Stallings

252 CHAPTER 5 / CONCURRENCY: MUTUAL EXCLUSION AND SYNCHRONIZATION

- 5.4 List three degrees of awareness between processes and briefly define each.
- 5.5 What is the distinction between competing processes and cooperating processes?
- 5.6 List the three control problems associated with competing processes and briefly define each.
- 5.7 List the requirements for mutual exclusion.
- 5.8 What operations can be performed on a semaphore?
- 5.9 What is the difference between binary and general semaphores?
- 5.10 What is the difference between strong and weak semaphores?
- 5.11 What is a monitor?
- 5.12 What is the distinction between *blocking* and *nonblocking* with respect to messages?
- 5.13 What conditions are generally associated with the readers/writers problem?

Module	Topic	Book
Module-2	Deadlock	Operating System- William Stallings

Review Questions

- 6.1 Give examples of reusable and consumable resources.
- 6.2 What are the three conditions that must be present for deadlock to be possible?
- 6.3 What are the four conditions that create deadlock?
- 6.4 How can the hold-and-wait condition be prevented?
- 6.5 List two ways in which the no-preemption condition can be prevented.
- 6.6 How can the circular wait condition be prevented?
- 6.7 What is the difference among deadlock avoidance, detection, and prevention?

available			
r1	r2	r3	r4
2	1	0	0

process	current allocation				maximum demand				still needs			
	r1	r2	r3	r4	r1	r2	r3	r4	r1	r2	r3	r4
p1	0	0	1	2	0	0	1	2				
p2	2	0	0	0	2	7	5	0				
p3	0	0	3	4	6	6	5	6				
p4	2	3	5	4	4	3	5	6				
p5	0	3	3	2	0	6	5	2				

- Compute what each process still might request and display in the columns labeled "still needs."
- Is this system currently in a safe or unsafe state? Why?
- Is this system currently deadlocked? Why or why not?
- Which processes, if any, are or may become deadlocked?
- If a request from p3 arrives for (0, 1, 0, 0), can that request be safely granted immediately? In what state (deadlocked, safe, unsafe) would immediately granting that whole request leave the system? Which processes, if any, are or may become deadlocked if this whole request is granted immediately?

6.6 Apply the deadlock detection algorithm to the following data and show the results.

$$\text{Available} = (2 \ 1 \ 0 \ 0)$$

$$\text{Request} = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 \end{pmatrix} \quad \text{Allocation} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 2 & 0 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{pmatrix}$$

Study Material Questions for Module II

Q.No.	Question	Blooms Level	CO	Marks																																																																																											
1	P1() { C=B-1; B=2*C; } P2() { D=2*B; B=D-1; } Here, B is a shared variable with initial value 2. How many values B can have? And what are the values?	4	2	10																																																																																											
2	A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e. wait) on a counting semaphore S and invokes the V operation (i.e. signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?	5	2	10																																																																																											
3	Does presence of cycle in a resource allocation graph necessarily creates deadlock. Explain.	6	2	10																																																																																											
4	Prove that deadlock prevention mechanism actually to prevent deadlock.	5	2	10																																																																																											
5	Consider the following snapshot of a system. There are no outstanding unsatisfied requests for resources. Check whether the system is in deadlock. <table><tr><th></th><th colspan="4">Current allocation</th><th colspan="4">Maximum demand</th><th colspan="4">Available</th></tr><tr><th>process</th><th>R1</th><th>R2</th><th>R3</th><th>R4</th><th>R1</th><th>R2</th><th>R3</th><th>R4</th><th>R1</th><th>R2</th><th>R3</th><th>R4</th></tr><tr><td>P1</td><td>0</td><td>0</td><td>1</td><td>2</td><td>0</td><td>0</td><td>1</td><td>2</td><td>1</td><td>5</td><td>2</td><td>0</td></tr><tr><td>P2</td><td>2</td><td>0</td><td>0</td><td>0</td><td>2</td><td>7</td><td>5</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td>0</td><td>0</td><td>3</td><td>4</td><td>6</td><td>6</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td>2</td><td>3</td><td>5</td><td>4</td><td>4</td><td>3</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P5</td><td>0</td><td>3</td><td>3</td><td>2</td><td>0</td><td>6</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr></table>		Current allocation				Maximum demand				Available				process	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4	P1	0	0	1	2	0	0	1	2	1	5	2	0	P2	2	0	0	0	2	7	5	0					P3	0	0	3	4	6	6	5	6					P4	2	3	5	4	4	3	5	6					P5	0	3	3	2	0	6	5	2					4	2	10
	Current allocation				Maximum demand				Available																																																																																						
process	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4																																																																																			
P1	0	0	1	2	0	0	1	2	1	5	2	0																																																																																			
P2	2	0	0	0	2	7	5	0																																																																																							
P3	0	0	3	4	6	6	5	6																																																																																							
P4	2	3	5	4	4	3	5	6																																																																																							
P5	0	3	3	2	0	6	5	2																																																																																							
6	Consider method used by process P1 and P2 for accessing critical section. The initial values of shared Boolean variables S1 and S2 are randomly selected. P1() { While (S1==S2); critical section S1=S2; } P2() { While(S1!=S2); critical section S1=not (S2); } Which of the following is true? Give explanation. a. Mutual exclusion + Progress b. Mutual exclusion only c. Progress only d. None	5	2	10																																																																																											

7	Consider method used by process P1 and P2 for accessing critical section. <pre> P1() P2() { { While (T) While(T) var P=T ; var Q=T while(var Q==T); while(var P==T); critical section critical section var P=F; var Q=F; } } </pre> Which of the following is true? Give explanation. - No Mutual exclusion + No Deadlock - Mutual exclusion only - Deadlock only - Mutual exclusion + Deadlock	5	2	10
8	A counting semaphore S is initialized to 10. Then, 6 P(Wait) operations and 4 V(Signal) operations are performed on S. What is the final value of S? Show the working.	5	2	5
9	“If there is a cycle in the resource allocation graph, it may or may not be in deadlock state“. Comment on this statement with a suitable example.	5	2	10

Subject	Operating System
Module	III
Chapters	7,8
Author	William Stallings

336 CHAPTER 7 / MEMORY MANAGEMENT

7.8 KEY TERMS, REVIEW QUESTIONS, AND PROBLEMS

Key Terms

absolute loading buddy system compaction dynamic linking dynamic partitioning dynamic run-time loading external fragmentation fixed partitioning frame internal fragmentation	linkage editor linking loading logical address logical organization memory management page page table paging partitioning	physical address physical organization protection relative address relocatable loading relocation segmentation sharing
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Review Questions

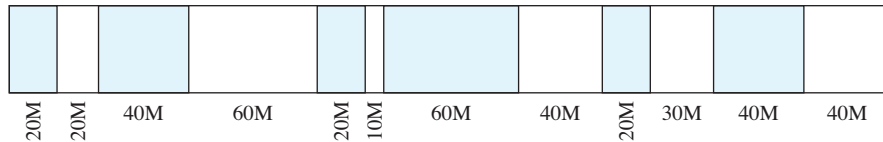
- 7.1 What requirements is memory management intended to satisfy?
- 7.2 Why is the capability to relocate processes desirable?
- 7.3 Why is it not possible to enforce memory protection at compile time?
- 7.4 What are some reasons to allow two or more processes to all have access to a particular region of memory?
- 7.5 In a fixed-partitioning scheme, what are the advantages of using unequal-size partitions?
- 7.6 What is the difference between internal and external fragmentation?
- 7.7 What are the distinctions among logical, relative, and physical addresses?
- 7.8 What is the difference between a page and a frame?
- 7.9 What is the difference between a page and a segment?

Problems

- 7.1 In Section 2.3, we listed five objectives of memory management, and in Section 7.1, we listed five requirements. Argue that each list encompasses all of the concerns addressed in the other.
- 7.2 Consider a fixed partitioning scheme with equal-size partitions of 2^{16} bytes and a total main memory size of 2^{24} bytes. A process table is maintained that includes a pointer to a partition for each resident process. How many bits are required for the pointer?
- 7.3 Consider a dynamic partitioning scheme. Show that, on average, the memory contains half as many holes as segments.
- 7.4 To implement the various placement algorithms discussed for dynamic partitioning (Section 7.2), a list of the free blocks of memory must be kept. For each of the three methods discussed (best-fit, first-fit, next-fit), what is the average length of the search?
- 7.5 Another placement algorithm for dynamic partitioning is referred to as worst-fit. In this case, the largest free block of memory is used for bringing in a process. Discuss the pros and cons of this method compared to first-, next-, and best-fit. What is the average length of the search for worst-fit?

7.8 / KEY TERMS, REVIEW QUESTIONS, AND PROBLEMS 337

- 7.6** A dynamic partitioning scheme is being used, and the following is the memory configuration at a given point in time:



The shaded areas are allocated blocks; the white areas are free blocks. The next three memory requests are for 40M, 20M, and 10M. Indicate the starting address for each of the three blocks using the following placement algorithms:

- First-fit
 - Best-fit
 - Next-fit. Assume the most recently added block is at the beginning of memory.
 - Worst-fit
- 7.7** A 1-Mbyte block of memory is allocated using the buddy system.
- Show the results of the following sequence in a figure similar to Figure 7.6: Request 70; Request 35; Request 80; Return A; Request 60; Return B; Return D; Return C.
 - Show the binary tree representation following Return B.
- 7.8** Consider a buddy system in which a particular block under the current allocation has an address of 011011110000.
- If the block is of size 4, what is the binary address of its buddy?
 - If the block is of size 16, what is the binary address of its buddy?
- 7.9** Let $\text{buddy}_k(x)$ = address of the buddy of the block of size 2^k whose address is x . Write a general expression for $\text{buddy}_k(x)$.
- 7.10** The Fibonacci sequence is defined as follows:

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+2} = F_{n+1} + F_n, \quad n \geq 0$$

- Could this sequence be used to establish a buddy system?
 - What would be the advantage of this system over the binary buddy system described in this chapter?
- 7.11** During the course of execution of a program, the processor will increment the contents of the instruction register (program counter) by one word after each instruction fetch, but will alter the contents of that register if it encounters a branch or call instruction that causes execution to continue elsewhere in the program. Now consider Figure 7.8. There are two alternatives with respect to instruction addresses:
- Maintain a relative address in the instruction register and do the dynamic address translation using the instruction register as input. When a successful branch or call is encountered, the relative address generated by that branch or call is loaded into the instruction register.
 - Maintain an absolute address in the instruction register. When a successful branch or call is encountered, dynamic address translation is employed, with the results stored in the instruction register.

Which approach is preferable?

- 7.12** Consider a simple paging system with the following parameters: 2^{32} bytes of physical memory; page size of 2^{10} bytes; 2^{16} pages of logical address space.
- How many bits are in a logical address?
 - How many bytes in a frame?
 - How many bits in the physical address specify the frame?
 - How many entries in the page table?
 - How many bits in each page table entry? Assume each page table entry contains a valid/invalid bit.

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page into each page frame, the time of last access to the page in each page frame, the virtual page number in each page frame, and the referenced (R) and modified (M) bits for each page frame are as shown (the times are in clock ticks from the process start at time 0 to the event — not the number of ticks since the event to the present).

Virtual page number	Page frame	Time loaded	Time referenced	R bit	M bit
2	0	60	161	0	1
1	1	130	160	1	0
0	2	26	162	1	0
3	3	20	163	1	1

A page fault to virtual page 4 has occurred at time 164. Which page frame will have its contents replaced for each of the following memory management policies? Explain why in each case.

- FIFO (first-in-first-out)
- LRU (least recently used)
- Clock
- Optimal (Use the following reference string.)
- Given the aforementioned state of memory just before the page fault, consider the following virtual page reference string:

4, 0, 0, 0, 2, 4, 2, 1, 0, 3, 2

How many page faults would occur if the working set policy with LRU were used with a window size of 4 instead of a fixed allocation? Show clearly when each page fault would occur.

- 8.5** A process references five pages, A, B, C, D, and E, in the following order:

A; B; C; D; A; B; E; A; B; C; D; E

Assume that the replacement algorithm is first-in-first-out and find the number of page transfers during this sequence of references starting with an empty main memory with three page frames. Repeat for four page frames.

- 8.6** A process contains eight virtual pages on disk and is assigned a fixed allocation of four page frames in main memory. The following page trace occurs:

1, 0, 2, 2, 1, 7, 6, 7, 0, 1, 2, 0, 3, 0, 4, 5, 1, 5, 2, 4, 5, 6, 7, 6, 7, 2, 4, 2, 7, 3, 3, 2, 3

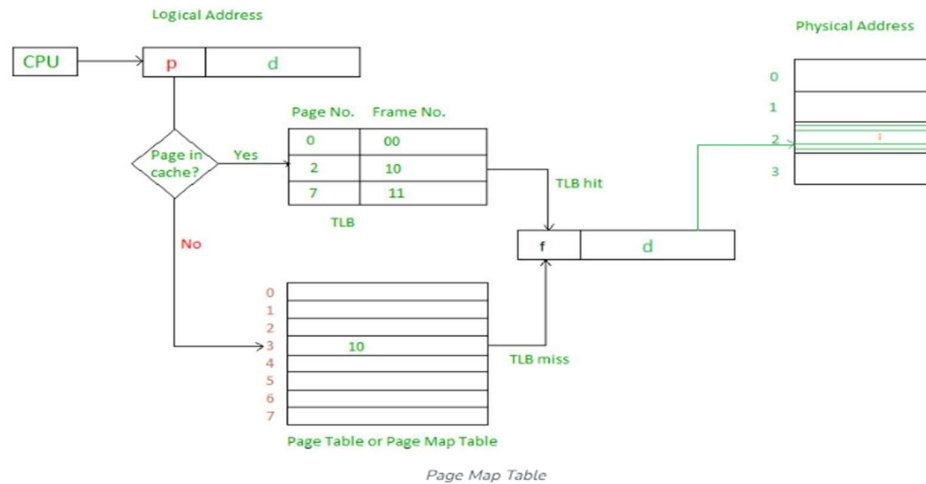
- Show the successive pages residing in the four frames using the LRU replacement policy. Compute the hit ratio in main memory. Assume that the frames are initially empty.
- Repeat part (a) for the FIFO replacement policy.
- Compare the two hit ratios and comment on the effectiveness of using FIFO to approximate LRU with respect to this particular trace.

- 8.7** In the VAX, user page tables are located at virtual addresses in the system space. What is the advantage of having user page tables in virtual rather than main memory? What is the disadvantage?

- 8.8** Suppose the program statement

```
for (i = 1; i <= n; i++)
    a[i] = b[i] + c[i];
```

Study Material Questions for Module III



Q.No.	Question	Blooms Level	CO	Marks												
1	"Consider the requests from processes in given order 300K, 25K, 125K, and 50K. Let there be two blocks of memory available of size 150K followed by a block size 350K. Which partition allocation schemes can satisfy the above requests? (NOTE : Here we assumed a partition can be allocated to a process even if some other process occupies a part of that partition.) Show the working neatly with diagram. Justify your answer."	6	3	5												
2	Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9 and 12 in that order. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves, to satisfy all the pending requests, for each of the following disk scheduling algorithms: FCFS, SSTF, SCAN & LOOK?	5	3	10												
3	i. Consider the reference string 6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0 for a memory with three frames and calculate number of page faults and page hits by using FIFO (First In First Out) Page replacement algorithm ii. Consider the reference string 6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0 for a memory with three frames and calculate number of page faults and page hits by using LRU (Least Recently Used) Page replacement algorithm	5	3	10												
4	Consider the 3 processes, P1, P2 and P3 shown in the table. <table><tr><th>Process</th><th>Arrival time</th><th>Time Units Required</th></tr><tr><td>P1</td><td>0</td><td>5</td></tr><tr><td>P2</td><td>1</td><td>7</td></tr><tr><td>P3</td><td>3</td><td>4</td></tr></table>	Process	Arrival time	Time Units Required	P1	0	5	P2	1	7	P3	3	4	5	3	10
Process	Arrival time	Time Units Required														
P1	0	5														
P2	1	7														
P3	3	4														

	What will be the completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units)?															
5	Consider a single level paging scheme with a TLB. Assume no page fault occurs. It takes 20 ns to search the TLB and 100 ns to access the physical memory. If TLB hit ratio is 80%, the effective memory access time is _____ msec.	4	3	5												
6	Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.	4	3	5												
7	Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?	5	3	10												
8	Consider the following table of arrival time and burst time for three processes P0, P1 and P2. <table><tr><td>Process</td><td>Arrival time</td><td>Burst Time</td></tr><tr><td>P0</td><td>0 ms</td><td>9 ms</td></tr><tr><td>P1</td><td>1 ms</td><td>4 ms</td></tr><tr><td>P2</td><td>2 ms</td><td>9 ms</td></tr></table> The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?	Process	Arrival time	Burst Time	P0	0 ms	9 ms	P1	1 ms	4 ms	P2	2 ms	9 ms	4	3	10
Process	Arrival time	Burst Time														
P0	0 ms	9 ms														
P1	1 ms	4 ms														
P2	2 ms	9 ms														
9	Assume every process requires 3 seconds of service time in a system with single processor. If new processes are arriving at the rate of 10 processes per minute, then estimate the fraction of time CPU is busy in system?	5		5												
10	Consider a logical address space of eight pages of 1024 words each, mapped onto a physical memory of 32 frames. a. How many bits are there in the logical address? b. How many bits are there in the physical address?	4		10												

Subject	Operating System
Module	IV
Chapters	12
Author	Stallings

12.14 KEY TERMS, REVIEW QUESTIONS, AND PROBLEMS

Key Terms

access method	file allocation	inode
bit table	file allocation table	key field
block	file directory	pathname
chained file allocation	file management system	pile
contiguous file allocation	file name	record
database	hashed file	sequential file
disk allocation table	indexed file	working directory
field	indexed file allocation	
file	indexed sequential file	

Review Questions

- 12.1** What is the difference between a field and a record?
- 12.2** What is the difference between a file and a database?
- 12.3** What is a file management system?
- 12.4** What criteria are important in choosing a file organization?
- 12.5** List and briefly define five file organizations.
- 12.6** Why is the average search time to find a record in a file less for an indexed sequential file than for a sequential file?
- 12.7** What are typical operations that may be performed on a directory?
- 12.8** What is the relationship between a pathname and a working directory?
- 12.9** What are typical access rights that may be granted or denied to a particular user for a particular file?
- 12.10** List and briefly define three blocking methods.
- 12.11** List and briefly define three file allocation methods.

Problems

- 12.1** Define:
 - B = block size
 - R = record size
 - P = size of block pointer
 - F = blocking factor; expected number of records within a block
 Give a formula for F for the three blocking methods depicted in Figure 12.8.
- 12.2** One scheme to avoid the problem of preallocation versus waste or lack of contiguity is to allocate portions of increasing size as the file grows. For example, begin with a portion size of one block, and double the portion size for each allocation. Consider a file of n records with a blocking factor of F , and suppose that a simple one-level index is used as a file allocation table.
 - a.** Give an upper limit on the number of entries in the file allocation table as a function of F and n .
 - b.** What is the maximum amount of the allocated file space that is unused at any time?

Study Material Questions for Module IV

Disadvantages of File System

- **Compatibility issues:** Different file systems may not be compatible with each other, making it difficult to transfer data between different operating systems.
- **Disk space overhead:** File systems may use some disk space to store metadata and other overhead information, reducing the amount of space available for user data.
- **Vulnerability:** File systems can be vulnerable to data corruption, malware, and other security threats, which can compromise the stability and security of the system.

Q.No.	Question	Blooms Level	CO	Marks
1	Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at cylinder 100, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105, 110, 135 and 145. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 90 is serviced after servicing which number of requests?	5	4	10
2	Consider an imaginary disk with 51 cylinders. A request comes in to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders 1, 36, 16, 34, 9 and 12 in that order. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves, to satisfy all the pending requests, for each of the following disk scheduling algorithms: FCFS, SSTF, SCAN & LOOK?	5	4	10
3	Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. What will be the additional distance (in terms of number of tracks) that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution)	5	4	10
4	Evaluate the efficiency and effectiveness of different file organization methods (e.g., sequential, indexed, hashed) in managing large volumes of data.	5	4	10
5	Synthesize a comprehensive file management strategy for a multinational corporation with geographically dispersed offices, considering factors such as security, accessibility, and scalability.	6	4	10
6	Assess the trade-offs between different disk scheduling algorithms (e.g., FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK) in terms of throughput, response time, and fairness, and recommend the most suitable algorithm for specific system configurations.	5	4	10